ALLSEIS-1C Neo
ALLSEIS-1C NeoHR
Seismic Nodal System

"ALLSEIS ®

— Unleashing the Power of Precision Seismic Imaging"

Geophysical Deep Sensing Technology

www.geodeepsensing.com

Product Description

ALLSEIS® - High Performance seismic nodal system

The new generation of ALLSEIS-1C Neo series nodal system is designed by GDST for higher resolution, higher efficiency, and higher reliability seismic acquisition with less operational cost and longer battery life.

This innovative nodal system is equipped with state-of-the-art sensors and seismic recording circuitry, providing reliable and accurate seismic signal digitalization results with unrivaled level of flexibility and quality control in the field.

With its light, compact, and rugged design, ALLSEIS-1C Neo Neo series seismic nodal system is the ideal solution for a wide range of geophysical applications such as energy and mineral exploration, urban geological structure imaging, CCUS, and passive seismic monitoring etc.

ABOUT GDST

Geophysical Deep Sensing Technology (GDST) is a leading designer and manufacturer of state-of-the-art smart sensors and seismic recording systems. GDST provides innovative solutions for energy and mineral exploration, urban geological structure imaging, and passive seismic monitoring applications.

GDST is fully committed to their nodal offering, ALLSEIS®, which provides a solution that caters to all needs with minimal environmental footprint in virtually any terrain. GDST currently serves more than 200 clients and operates over 300,000 channels for both seismic and industrial operations.

Unrivaled Sensor Flexibility

ALLSEIS-1C Neo provides four different formfactors that can accommodate different seismic acquisition scenarios. The standard form integrates a high sensitivity geophone (5Hz, 10Hz, or others upon request). And the KCK form supports a variety of external geophone strings, such as 20DX_6S1P, 20DX_6S2P, SM24_6S1P, SM24_6S2P, 32CT_6S1P, 32CT_6S2P, etc. Special requirements can also be supported according to customer demand. ALLSEIS-1C Neo features automatic built-in test function which supports on-site full signal chain function and performance self-inspection and diagnosis, including system noise, dynamic range, geophone impedance, natural frequency, sensitivity, damping, distortion, etc..

Best-in-class Downloading and Charging Speed

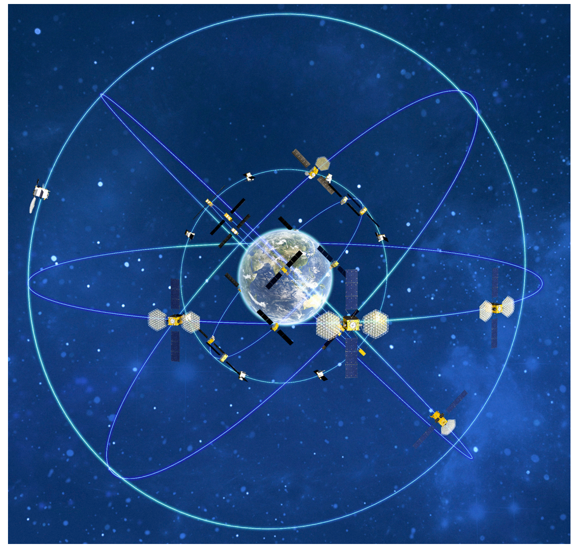
One ALLSEIS-1C Neo node generates 4 Giga Bytes of raw data after 20 days of continuous recording at 500 SPS, which can be downloaded in 3 minutes at a single slot of the 32-slot standard downloading rack. Two downloading racks can handle at least 500 such nodes per hour, corresponding to over 2 Tera Byte of raw data.

GDST also provides multiple choices of auxiliary devices for different application scenarios: the 32-slot downloading rack and charging rack are suitable for large-scale seismic survey running for months. For medium-scale seismic survey using thousands of nodes, the 14-slot desktop downloading rack might be appropriate.

Dual GNSS Synchronization Strategy

ALLSEIS-1C Neo node is equipped with a high-sensitivity satellite positioning and timing module that supports multi-satellite GNSS constellation including GPS, Beidou, GLONASS, and others.

ALLSEIS-1C Neo node features a unique dual-mode GNSS clock synchronization strategy to achieve a balance between GNSS-disciplined timing accuracy and battery power consumption. In instances where the GNSS signal strength is strong, the node operates in intermittent synchronization mode to conserve batteries. If the GNSS signal strength weakens because of sheltering, burying, etc., the node will automatically switch to continuous synchronization mode to provide the best possible clock synchronization.



Long Range BLE for Field QC and Stakeless Deployment

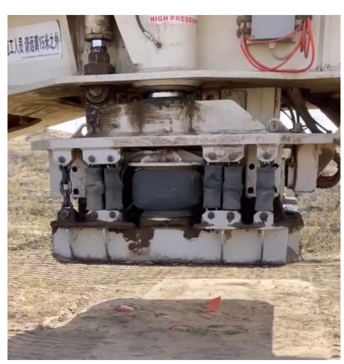
ALLSEIS-1C Neo node is equipped with BLE wireless module that enables field personnel, vehicle, or drone to patrol for node status verification as well as real-time seismic wave inspection.

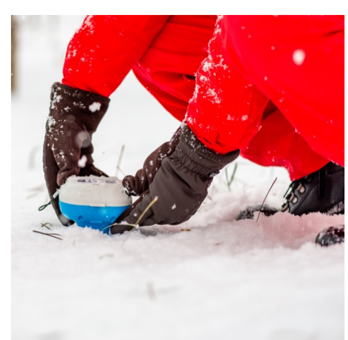
Field QC information, including sampling parameters, storage status, battery autonomy, temperature, tilting, ambient noise, etc., assists crew personnel easily deploying, troubleshooting and retrieving ground equipment. The BLE antenna is enhanced for long range data transmission, exceeds 20 meters on the ground or 100 meters in the air.

ALLSEIS-1C Neo node also features P&R (Place & Record) mode that automatically matches node's GNSS position with line and point number for stakeless surveying.

Sturdy industrial design

ALLSEIS-1C Neo node is well known for its robust construction. It is built with high strength engineering plastic enclosure to enhance its durability and overall performance from rough handling during deployment and retrieval in harsh environmental conditions, such as, extreme temperatures, ultraviolet radiation, salt atmosphere, etc.





General Parameter of ALLSEIS-1C Neo Series Nodes

Sensor	5Hz High Sensitivity Geophone (5Hz~150Hz) 10Hz High Sensitivity Geophone (10Hz~240Hz) Optional KCK Socket Version for External String
Digitalization	32-bit Δ-∑ ADC with GNSS disciplined clock
Storage Capacity	16GB (32GB optional)
Node Autonomy	Internal 54 lithium-ion battery pack supports 1080/1200 hours continuous or intermittent recording
Timing Accuracy	+/- 5μs (continuous GNSS disciplined) +/- 1ms (within 1 hour without GNSS signal)
Automated Tests	Internal testing signal generator for on-site full signal chain function and performance self-inspection
Working Mode	Autonomy with BLE wireless QC inspection Support UAV patrol
LED Status Indicator	Nodal health, GNSS synchronization status, BLE wireless data transmission status
Data Harvest	Charging / Data download rack
Infield Management	Handheld android tablet
Weight	750g (excl. spike), 800g (incl. spike)
Size	9.8cm x 10.8cm x 11cm (Excl. spike)
Operating Temperature	-40°C ~ +70°C

** Specifications or information subject to change without notice*

ALLSEIS-1C Neo

ALLSEIS-1C NeoHR w/ KCK

Acquisition Specifications

Item	ALLSEIS-1C Neo	ALLSEIS-1C NeoHR
ADC Resolution	32-bit	32-bit
Sample Rates	0.5ms, 1ms, 2ms, 4ms	0.25ms, 0.5ms, 1ms, 2ms, 4ms
Preamplifier Gain	x1, x2, x4, x8, x16, x32, x64 (0dB, 6dB, 12dB, 18dB, 24dB, 30dB, 36dB)	
Input Impedance	40KΩ || 22nF (5Hz geophone) 20KΩ || 22nF (10Hz geophone)	
Gain Accuracy	0.1%	
Full Scale Input Signal	±2.5Vp-p@x1 gain, ±625mVp-p@x4 gain ±156mVp-p@x16 gain, ±39mVp-p@x64 gain	
Instantaneous DR @4ms (typical value)	125dB @ x1 gain 122dB @ x4 gain 115dB @ x16 gain 106dB @ x64 gain	130dB @ x1 gain 128dB @ x4 gain 123dB @ x16 gain 114dB @ x64 gain
Full Scale DR	142dB	150dB
Equivalent Input Noise @4ms (typical value)	0.99μV @ x1 gain 0.35μV @ x4 gain 0.20μV @ x16 gain 0.17μV @ x64 gain	0.50μV @ x1 gain 0.15μV @ x4 gain 0.08μV @ x16 gain 0.07μV @ x64 gain
Harmonic Distortion	-120dB @ 31.5Hz	
Anti-alias Filter	Linear or minimum phase filter, 82.6% of Nyquist -3dB bandwidth	Linear or minimum phase filter, 82.6% of Nyquist -3dB bandwidth
Frequency Response @1ms	0Hz ~ 413Hz (-3dB)	
Programmable HPF	0.1~10 Hz	
Stopband Attenuation	>130dB @Nyquist	
Common Mode Rejection	>120dB	

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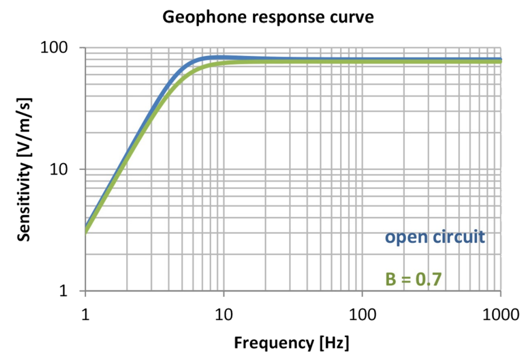
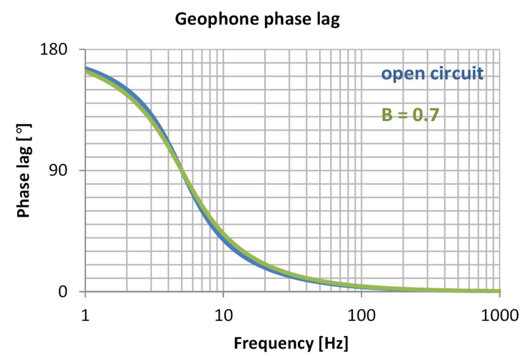
Geophone Specifications

5Hz High Sensitivity Geophone

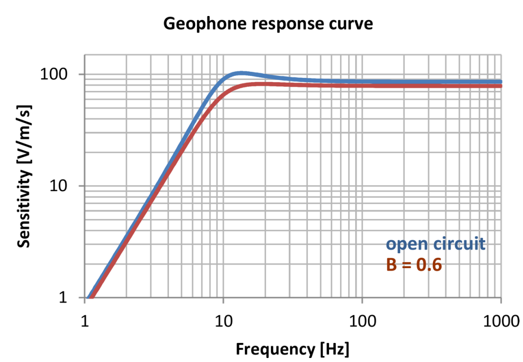
Resistance	
Coil resistance	1850 Ω
Tolerance	± 5 %
Frequency	
Natural frequency(fn)	5Hz
Tolerance	± 7.5 %
Typical spurious frequency	> 150 Hz
Damping	
Open –circuit damping	0.6
Tolerance	± 7.5 %
Sensitivity	
Open-circuit sensitivity	80V/m/s (2.03 V/in/s)
Tolerance	± 5 %
Distortion	
Distortion with 0.7 V/in/s p.p. coil-to-case velocity	0.1 % max
Distortion measurement frequency	12 Hz
Maximum tilt angle for specified distortion	10°
RtBcFn	22436 Ω Hz
Moving mass	22.7 g (0.8 oz)
Maximum coil excursion p.p.	3 mm (0.12 in)
Physical Characteristics	
Diameter	30.5 mm (1.2 in)
Height	41.5 mm (1.63 in)
Weight	140 g (4.94 oz)
Operating temperature	-40~100 °C (-40~212 °F)

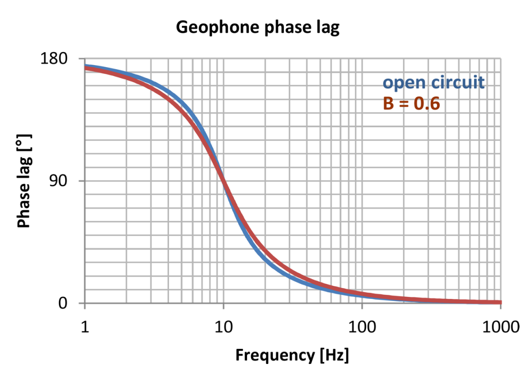




10Hz High Sensitivity Geophone

Resistance	
Coil resistance	1800 Ω
Tolerance	± 3.5 %
Frequency	
Natural frequency(fn)	10Hz
Tolerance	± 3.5 %
Typical spurious frequency	> 240 Hz
Damping	
Open –circuit damping	0.475
Damping with 20 kΩ system input impedance	0.7
Tolerance with 20 kΩ system input impedance	± 3.5 %
Sensitivity	
Open-circuit sensitivity	85.8V/m/s (2.08V/in/s)
Tolerance	± 3.5 %
Distortion	
Distortion with 0.7 V/in/s p.p. coil-to-case velocity	<0.2 %
Distortion measurement frequency	12 Hz
Maximum tilt angle for specified distortion	20°
Moving mass	11.8 g (0.42 oz)
Maximum coil excursion p.p.	2 mm (0.08 in)
Physical Characteristics	
Diameter	26.7 mm (1.05 in)
Height	32 mm (1.26 in)
Weight	85.6 g (3.02 oz)
Operating temperature	-40~100 °C (-40~212 °F)





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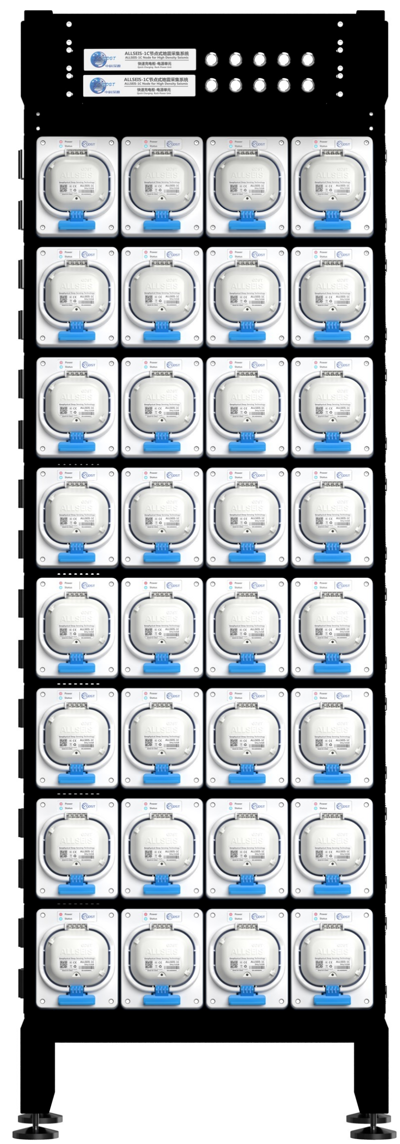
System Components



STANDARD DOWNLOAD RACK

- Capacity: 32 Slots.
- Unit download rate: 0.6 GB/min.
- Full download rate: 18 GB/min.
- Interface: 10G SFP+.
- Operating temperature: 0°C to +40°C.
- Dimensions: 194 x 64 x 38 cm

STANDARD CHARGING RACK

- Capacity: 32 Slots.
- Charing time: 3.5 Hours
- Charing power: 1500W
- Operating temperature: 0°C to +40°C.
- Dimensions: 194 x 64 x 38 cm.

INTEGRATED DOWNLOAD/CHARGING RACK

- Capacity: 32 Slots.
- Charing time: 3.5 Hours
- Charing power: 1500W
- Unit download rate: 0.6 GB/min.
- Full download rate: 18 GB/min.
- Interface: 10G SFP+.
- Operating temperature: 0°C to +40°C.
- Dimensions: 194 x 64 x 38 cm.

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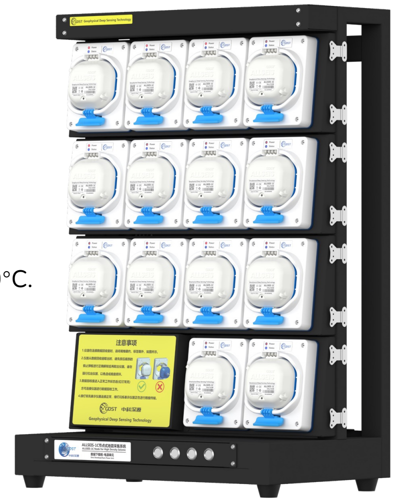
System Components

DESKTOP DOWNLOAD/CHARGING RACK

- Capacity : 14 Slots.
- Charing time: 3.5 Hours
- Charing power: 700W
- Unit download rate : 0.6 GB/min.
- Full download rate : 8 GB/min.
- Interface : 2.5Gb Ethernet.
- Operating temperature : 0°C to +40°C.
- Dimension : 94x 64x 38 cm



PORTABLE DOWNLOAD UNIT

- Download speed : 1.2 GB/min.
- Interface : USB 2.0.
- Operating temperature : 0°C to +40°C.

PORTABLE DOWNLOAD RACK

- Capacity : 3 Slots.
- Unit download rate : 0.6 GB/min.
- Full download rate : 1.8 GB/min.
- Interface : 1Gb Ethernet.
- Operating temperature : 0°C to +40°C.
- Dimension : 22 x 49 x 15 cm

PORTABLE CHARGING RACK

- Capacity : 9 Slots.
- Charing time : 3.5 Hours
- Charing power : 200W
- Operating temperature : 0°C to +40°C.
- Dimension : 65x 54x 30 cm.

TIME BREAK RECORDER

- Record TB pulse or closed signal excitation on Android App.
- Timing Accuracy : ±1us
- Internal Battery Life : 60 hours
- Operating temperature : -40°C ~ +70°C
- Water Proof : IP67

Software Suite

ALLSEIS software suite is composed with the SeisData Management (SDM) software, the In-field QC App., and other software tools. It covers all necessary aspects of source-driven seismic acquisition, including nodal configuration, equipment deployment and retrieval, operational QC and reporting, seismic data download, creation of shot gather and receiver gather, vibroseis correlation and stacking, seismic data quality control, etc. It is a comprehensive solution for managing data archives and quality control in hybrid or standalone nodal operations.

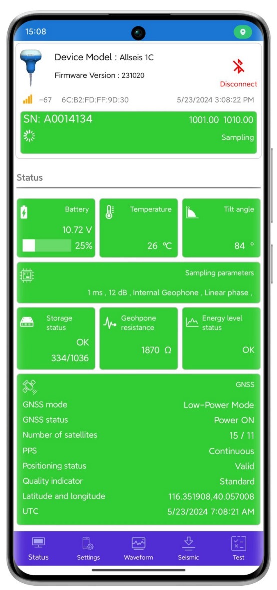
Key features of the software include:

- Nodal script configuration
- Function and performance diagnosis for nodes
- Management of node deployment and retrieval
- Seismic data download, sorting, and merging
- Creation of shot gather and receiver gather
- Vibroseis correlation and stacking
- Output in SEG-Y or SEG-D format
- Combining traces for hybrid cable/nodal systems
- Calculation of seismic trace yield
- Seismic data quality control
- Generation of graphical and tabular reports

◆ *The SDM server is recommended to equip with RAID array comprising Solid State Drives (SSDs) for high throughput data manipulation.*

The ALLSEIS-1C Neo in-field QC App. is an intuitive and user-friendly software application running on Android mobile phones or tablets that are built for harsh terrains. Such handhold QC tool can be used during node deployment and retrieval by field personnel, vehicle, or drone for navigation and path guidance, node status verification, as well as real-time seismic wave inspection.

Field QC information, including sampling parameters, storage status, battery autonomy, temperature, tilting, ambient noise, etc. can be accessed through the SDM software to generate graphical and/or tabular reports, so that further optimize crew productivity throughout the acquisition lifecycle.



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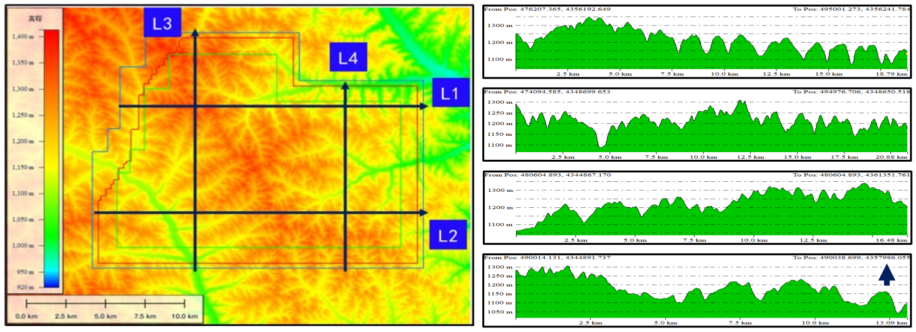
Case Study: Fugu County 3D Exploration Project

Project Overview:

- Running by CNOOC
- Typical loess tableland area
- Elevation : 984m - 1386m
- Elevation difference : 402m
- 3D area : 301.22 km^2
- Shot points : 21818
- Receiver points : 47692
- Receiver interval : 40m



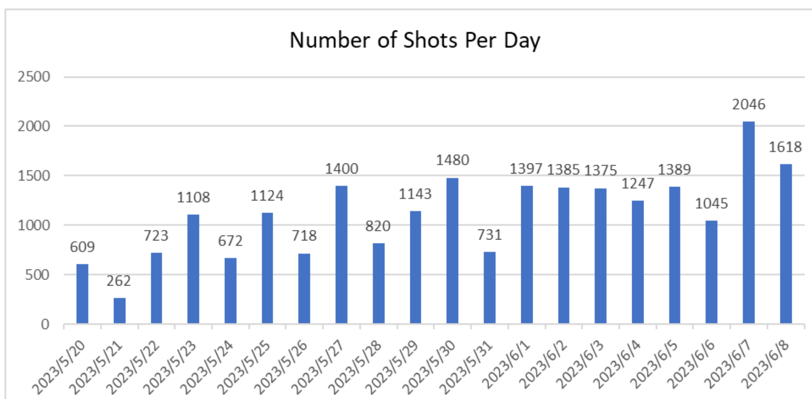
300Km² loess tableland area in north west part of China, covered by luxuriant vegetation, elevation difference over 400m

Equipment Utilization:

- ALLSEIS-1C : 30000
- Standard Download Rack : 4
- Standard Charging Rack : 25
- Download Server : 2
- SDM Server : 1
- BoomBox-III : 21
- UAV : 1

Operating Efficiency:

- Field Crews : 65
- Daily Deployed/ channels : ~2500
- Daily Retrieved channels : ~2500
- Daily UAV Patrol channels : ~3000
- Daily Download Raw Data : 9TB
- Daily shots : 1066 in average
- Project Duration : 21days
- Percentage of Bad Traces : < 0.3%



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